Theorem 6: the upper bound 2 for the edit distance

Proof overview of the Lean 4/Mathlib formalisation

Folder `GsimOT/Theorem6_EditUpperBound`

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport.*
File `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository; statement, equation, page and line numbers refer to this PDF. `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

1 Statement

Paper. Theorem 6 (Section 3.1, p. 7, l. 245–247; proof in Appendix C.2, pp. 24–25, l. 761–785): *Let* $Q \in \mathrm{DS}(V,W)$ *with a decomposition as in Eq. (3). Then* $\sum_{i=1}^{m} \lambda_i C_{\mathrm{ed}}(\pi_i) \le 2\, C_{\mathrm{ed}}(Q)$. *In particular,* $d_{\mathrm{ed}}(G,H) \le 2\, d_{\mathrm{ed}}^{\mathrm{OT}}(G,H)$ *for all graphs* G, H *of the same order.* Eq. (3) (p. 5, l. 193–195) is a Birkhoff decomposition $Q = \sum_{i=1}^{m} \lambda_i P_{\pi_i}$ with $\lambda_i > 0$ and $\sum_i \lambda_i = 1$. Normalised by n^2, the second statement is the first bound of Theorem 2 (p. 3, l. 119), $\delta_{\mathrm{ed}} \le 2\, \delta_{\mathrm{ed}}^{\mathrm{OT}}$.

Lean. For finite types V, W and simple graphs G on V, H on W:

- `GsimOT.theorem6`: for a finite index type ι, weights $\lambda\colon \iota \to \mathbb{R}$ with $\lambda_i \ge 0$ and $\sum_i \lambda_i = 1$, bijections $\pi_i\colon V \to W$, and Q with $Q(v,w) = \sum_i \lambda_i P_{\pi_i}(v,w)$,

$$\sum_i \lambda_i\, C_{\mathrm{ed}}(\pi_i) \le 2\, C_{\mathrm{ed}}(Q).$$

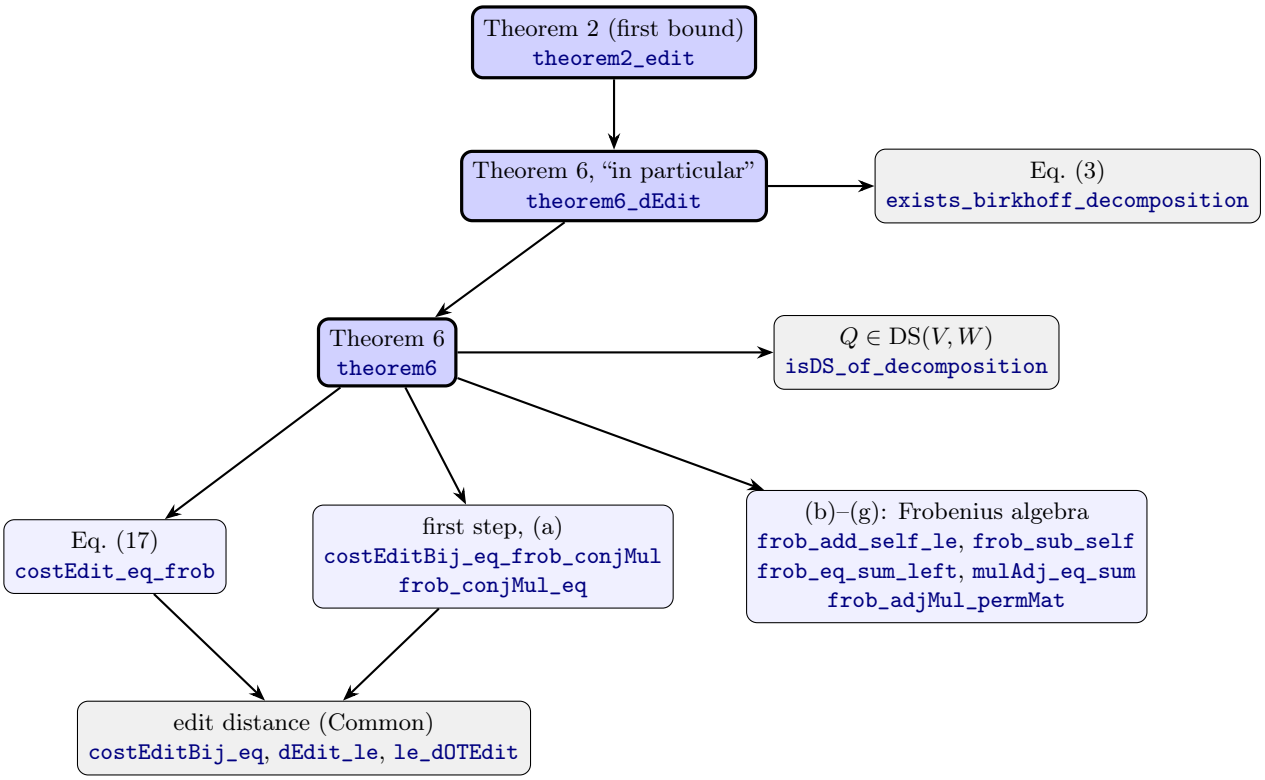
- `GsimOT.theorem6_dEdit`: $d_{\mathrm{ed}}(G,H) \le 2\, d_{\mathrm{ed}}^{\mathrm{OT}}(G,H)$, for all G, H.
- `GsimOT.theorem2_edit` (Theorem 2, first bound): $\delta_{\mathrm{ed}}(G,H) \le 2\, \delta_{\mathrm{ed}}^{\mathrm{OT}}(G,H)$, for all G, H.

The Lean statements are slightly stronger: zero weights are allowed, $Q \in \mathrm{DS}(V,W)$ is not assumed but follows (`isDS_of_decomposition`), and the second and third statements need no assumption $|V| = |W|$ (otherwise there is no bijection and $d_{\mathrm{ed}} = 0$). C_{ed}, d_{ed}, $d_{\mathrm{ed}}^{\mathrm{OT}}$ are `costEditBij`/`costEdit`, `dEdit`, `dOTEdit` of `Common/EditDistance.lean`, shared with Proposition 5 and Theorem 15.

2 Where the proof lives

File	Content
`TraceFormula.lean`	the Frobenius inner product `frob`, the products $A_G R$, $R A_H$, $R A_H R^{\top}$ (`adjMul`, `mulAdj`, `conjMul`), the algebraic facts used in the chain, and Eq. (17) (`costEdit_eq_frob`).
`Theorem6.lean`	Theorem 6 as a `calc` chain (`theorem6`), the "in particular" (`theorem6_dEdit`) and Theorem 2 for δ_{ed} (`theorem2_edit`).
`Common/BirkhoffDecomposition.lean`	Eq. (3): `exists_birkhoff_decomposition` (from Mathlib's Birkhoff–von Neumann theorem) and `isDS_of_decomposition`.
`Common/EditDistance.lean`	C_{ed}, d_{ed}, $d_{\mathrm{ed}}^{\mathrm{OT}}$, `costEditBij_eq`, `dEdit_le`, `le_dOTEdit`.

3 Proof layout



Arrows mean "uses". Blue: this folder; grey: `GsimOT/Common`. Nothing is cited; Eq. (3) is derived from Mathlib's `exists_eq_sum_perm_of_mem_doublyStochastic`.

4 The proof step by step

Setting (l. 761–762). The paper takes the random bijection π with $\Pr(\pi = \pi_i) = \lambda_i$ and $P := P_\pi$, so $\mathbb{E}[P] = Q$. The formalisation does not model randomness: $\mathbb{E}[f(\pi)]$ is the sum $\sum_i \lambda_i f(\pi_i)$. In `theorem6`, $X_i = A_G P_{\pi_i}$, $Y_i = P_{\pi_i} A_H$ and $C = QA_H$; the facts $\mathbb{E}[P] = Q$ are `mulAdj_eq_sum` ($QA_H = \sum_i \lambda_i P_{\pi_i} A_H$) and `adjMul_eq_sum` ($A_G Q = \sum_i \lambda_i A_G P_{\pi_i}$), and $\mathbb{E}[c] = c$ uses $\sum_i \lambda_i = 1$.

Eq. (17) (l. 762–765). $C_{\mathrm{ed}}(Q) = \|A_G\|_F^2 + \|A_H\|_F^2 - 2\langle A_G Q, QA_H\rangle_F$ for $Q \in \mathrm{DS}(V, W)$: `costEdit_eq_frob`. As in the paper, `abs_disc_eq` expands $|a - b| = a^2 + b^2 - 2ab$ for $a, b \in \{0, 1\}$, the row sums reduce the first term to $\|A_G\|_F^2$, the column sums the second to $\|A_H\|_F^2$, and the mixed term is regrouped into $\langle A_G Q, QA_H\rangle_F$.

The chain (pp. 24–25, l. 767–777). The `calc` block of `theorem6` has one step per line, commented with the paper's line and label:

Line, label	Paper	Lean
l. 767	$\mathbb{E}[C_{\mathrm{ed}}(\pi)] = \mathbb{E}\|A_G - PA_H P^\top\|_F^2$	`costEditBij_eq_frob_conjMul` (entries in $\{0,1\}$, so $\|a-b\| = (a-b)^2$; uses `costEditBij_eq`)
l. 768, (a)	permutation invariance: $\|A_G P - PA_H\|_F^2$	`frob_conjMul_eq`
l. 769	regrouping $(A_G P - QA_H) + (QA_H - PA_H)$	`sub_add_sub_cancel`
l. 770, (b)	$\|X + Y\|_F^2 \le 2\|X\|_F^2 + 2\|Y\|_F^2$	`frob_add_self_le`
l. 771, (c)	expanding the second square	`frob_sub_self`
l. 772, (d)	$\mathbb{E}\langle QA_H, PA_H\rangle_F = \langle QA_H, QA_H\rangle_F$	`frob_eq_sum_right`, `mulAdj_eq_sum`
l. 773, (e)	$\langle QA_H, QA_H\rangle_F = \|QA_H\|_F^2$, collecting terms	`ring`
l. 774, (f)	expanding the first square	`frob_sub_self`

Line, label	Paper	Lean
l. 775	cancelling $\|QA_H\|_F^2$	`ring`
l. 776, (g)	$\mathbb{E}[P] = Q$, permutation invariance	`frob_eq_sum_left`, `adjMul_eq_sum`, `frob_adjMul_permMat`, `frob_mulAdj_permMat`
l. 777, Eq. (17)	$= 2\,C_{\mathrm{ed}}(Q)$	`costEdit_eq_frob` with `isDS_of_decomposition`

The paper explains steps (a)–(g) in l. 778–782.

Second assertion (p. 25, l. 783–785). The paper takes a minimiser Q of C_{ed} over $\mathrm{DS}(V, W)$ and samples π from its decomposition: $d_{\mathrm{ed}} \le \mathbb{E}[C_{\mathrm{ed}}(\pi)] \le 2C_{\mathrm{ed}}(Q) = 2d_{\mathrm{ed}}^{\mathrm{OT}}$. As $d_{\mathrm{ed}}^{\mathrm{OT}}$ is defined as an infimum, `theorem6_dEdit` applies Theorem 6 to *every* $Q \in \mathrm{DS}(V, W)$, with the decomposition `exists_birkhoff_decomposition`: $d_{\mathrm{ed}} \le \sum_\sigma \lambda_\sigma C_{\mathrm{ed}}(\pi_\sigma) \le 2C_{\mathrm{ed}}(Q)$ (`dEdit_le`), and concludes with `le_dOTEdit`. `theorem2_edit` divides by n^2.

5 Assumptions and trusted base

No cited result and no hypothesis. The Birkhoff–von Neumann theorem comes from Mathlib. The main theorems depend only on `propext`, `Classical.choice`, `Quot.sound` (checked by `scripts/CheckAxioms.lean`).

6 Notes on the formalisation

- **Hypotheses.** Only $\lambda_i \ge 0$ and $\sum_i \lambda_i = 1$ are assumed, over any finite index type, so every decomposition as in Eq. (3) is covered; $Q \in \mathrm{DS}(V, W)$ follows (`isDS_of_decomposition`). The decomposition `exists_birkhoff_decomposition` used for the second assertion sums over all permutations of V, some with weight 0.
- **Infimum instead of minimiser** and **no assumption** $|V| = |W|$ in `theorem6_dEdit`, see above. (The infimum is attained, `exists_dOTEdit_eq`, but this proof does not need it.)
- **Not formalised**: Example 7 (p. 7, l. 250–259) and the closing remark of Appendix C.2 (p. 25, l. 786–789; arbitrary matrices via thresholds).

7 How to check this result

- Build: `lake build GsimOT.Theorem6_EditUpperBound.Theorem6` (or `bash build.sh` for the whole library, which also runs `scripts/CheckAxioms.lean`).
- Axioms: `#print axioms GsimOT.theorem6`, `GsimOT.theorem6_dEdit`, `GsimOT.theorem2_edit`, `GsimOT.costEdit_e` expected output: `propext`, `Classical.choice`, `Quot.sound`.
- The table `Theorem6_correspondence.pdf` in this folder lists every paper item with its Lean declaration and source location.